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AI for Bharat Hackathon

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Team Name : The Sentinels

Team Leader Name : Abhinav Khare

Problem Statement : Build an AI-powered solution that helps people learn faster, work smarter, or become more productive while building or understanding technology.

Brief about the Idea:

Code Guardian is an educational VS Code extension designed to intercept AI-generated code. Instead of blindly pasting, developers must explain the code logic in their own words. Our AI backend grades their understanding. If they fail three times, it switches to a supportive "Explain Like I'm 5" (ELI5) mentor mode.

Why AI is required: Traditional rule-based systems cannot evaluate if a student genuinely understands code. We need an LLM to semantically grade natural language explanations.

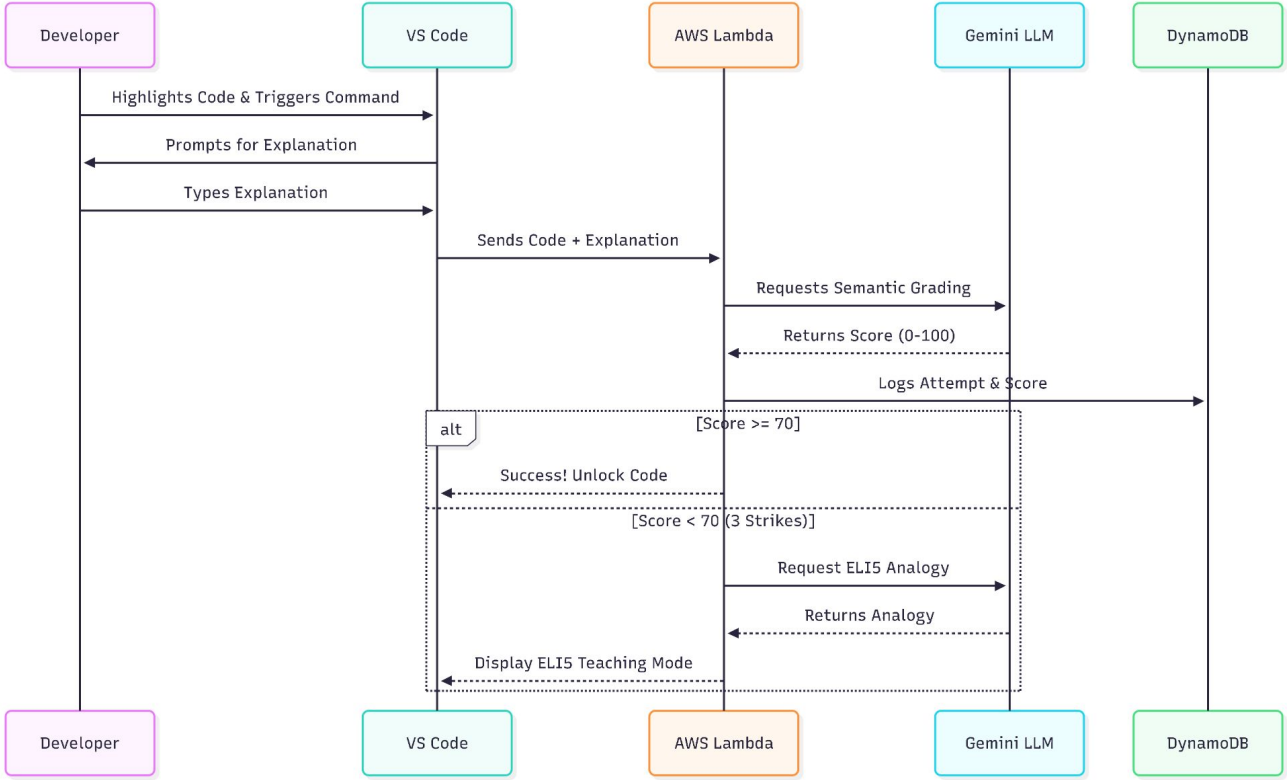
How AWS services are used: AWS Lambda runs the serverless Python backend to orchestrate the AI grading. Amazon DynamoDB persistently logs user session data, scores, and strike counts.

Value the AI layer adds: It transforms a "friction point" into a personalized educational mentor, preventing skill degradation.

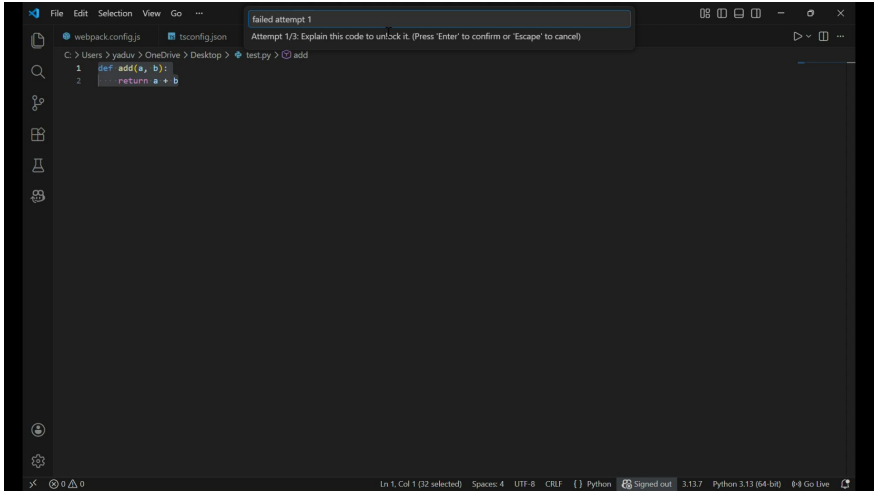
List of features offered by the solution:

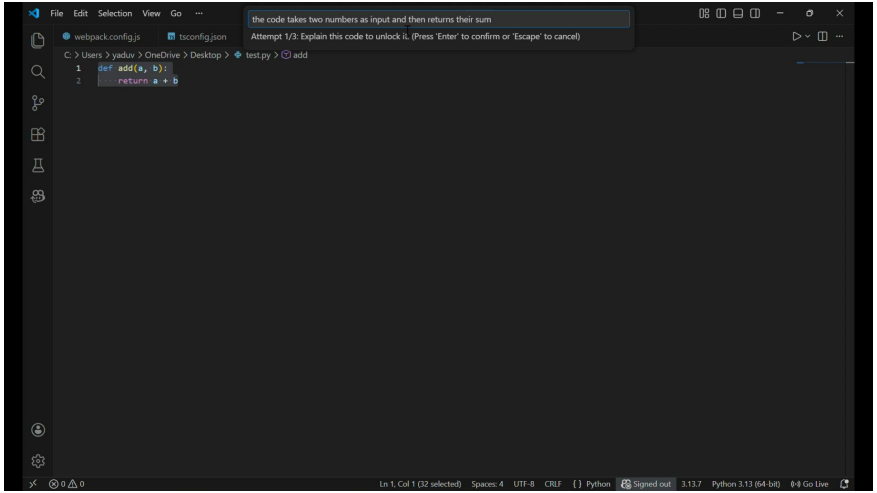
- **Real-Time Semantic Grading:** Scores explanations on a 0-100 scale.
- **The 70% Threshold:** Unlocks the developer's workflow only if they prove understanding.
- **"3-Strike" ELI5 Fallback:** Automatically simplifies complex logic using real-world analogies after 3 failed attempts.
- **Cloud-Based Tracking:** Logs all learning history in DynamoDB.

Process flow diagram or Use-case diagram

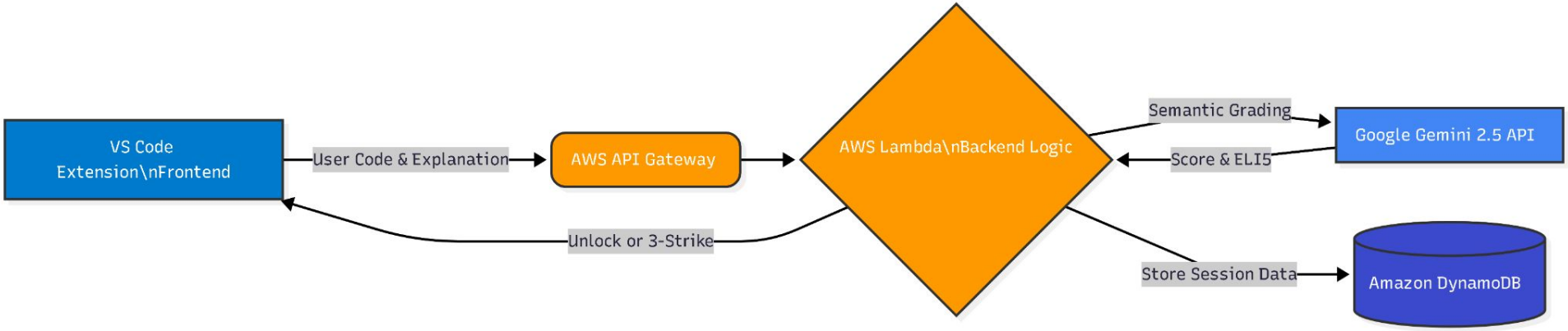


Wireframes/Mock diagrams of the proposed solution





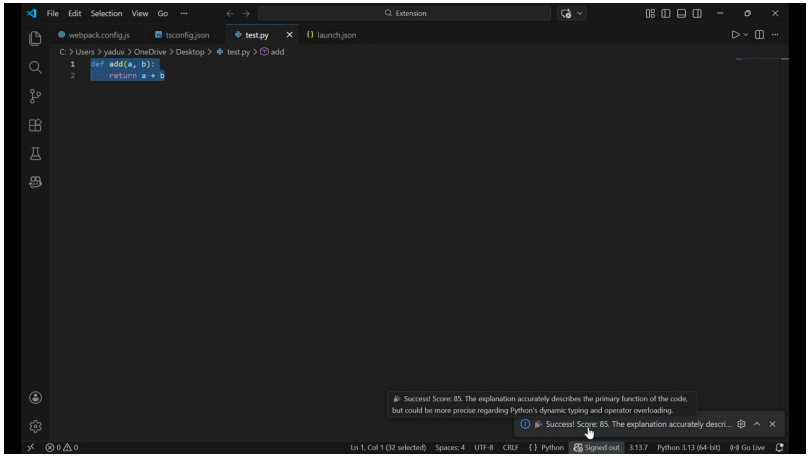
Architecture diagram of the proposed solution:

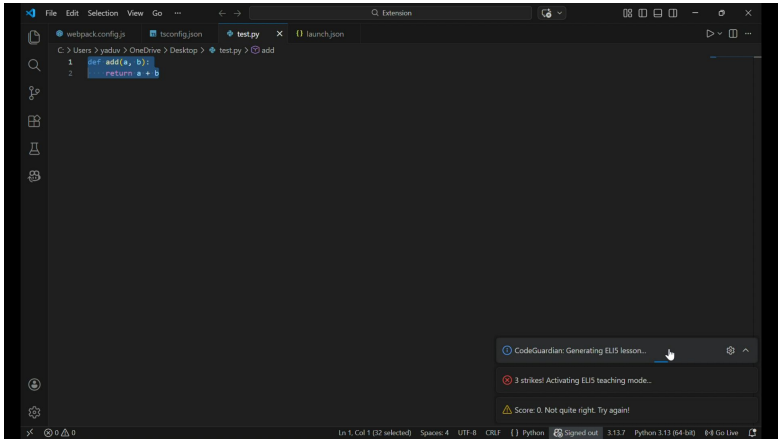


Technologies utilized in the solution:

- **Frontend:** Visual Studio Code Extension API, TypeScript, Node.js
- **Backend Cloud:** AWS Lambda, Amazon API Gateway
- **Database:** Amazon DynamoDB
- **AI/LLM Engine:** Google Gemini 2.5 Flash(For the MVP only (Actual amazon bedrock model claude sonnet 3.5 didnt work))
  [The final working extension will be using Amazon bedrock models.]

Snapshots of the prototype:





Prototype Performance report/Benchmarking:

Latency: Sub-2 second end-to-end response time from VS Code to AWS Lambda and back.

Uptime: High availability powered by AWS Serverless architecture.

Accuracy: Effectively evaluates logic using semantic meaning rather than basic keyword matching.

Additional Details/Future Development:

- **Auto-Paste Interception:** Hooking directly into VS Code's clipboard API (moving beyond the MVP highlight requirement).
- **Multilingual Support:** Explanations and ELI5 analogies in regional languages (Hindi, Tamil, etc.).
- **Voice-to-Explain:** Audio transcription for verbal explanations.
- **Gamification:** Earning knowledge points and daily streaks.
- **Interview Prep Mode:** Dynamically generated Q&A rounds based on pasted code.
- **Code Optimization:** Contextual AI tips (e.g., time complexity, library fun facts).

Prototype Assets:

- ➔ GitHub Public Repository Link:
- ➔ https://github.com/yaduvansh-kumra/CodeGuardian---AI-Hackathon
- ➔
- ➔ Demo Video Link :
  https://drive.google.com/drive/folders/1ojrNcGFhb4a3t9hZV0OZtstvsjZ9sqKD?usp=sharing

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Thank You